Project Design Phase

Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-3075
Project Name	Streamlining IT Procurement – Automating Standard Laptop Orders
Team Size	4
Team Leader	Karpagam R
Team Member	Padma Priya R
Team Member	Mariyam Asiya S
Team Member	Shreya S
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for **Streamlining IT Procurement – Automating Standard Laptop Orders** defines how ServiceNow can automate and manage the university's standard laptop procurement process. The solution connects employee requests, approval workflows, procurement activities, inventory information, and notifications into a single controlled process.

• Provide a simple ServiceNow request process for employees or students to order an approved standard laptop.

• Automatically validate the request against predefined laptop models, eligibility rules, and required information.

• Route requests through the appropriate approval workflow before procurement action is initiated.

• Create and track procurement tasks automatically, while keeping request, approval, delivery, and asset information linked.

• Send status notifications to the requester and relevant university IT/procurement teams at important stages.

• Maintain an auditable record of requests, approvals, procurement status, and assigned laptop assets.

Example – Solution Architecture Diagram:

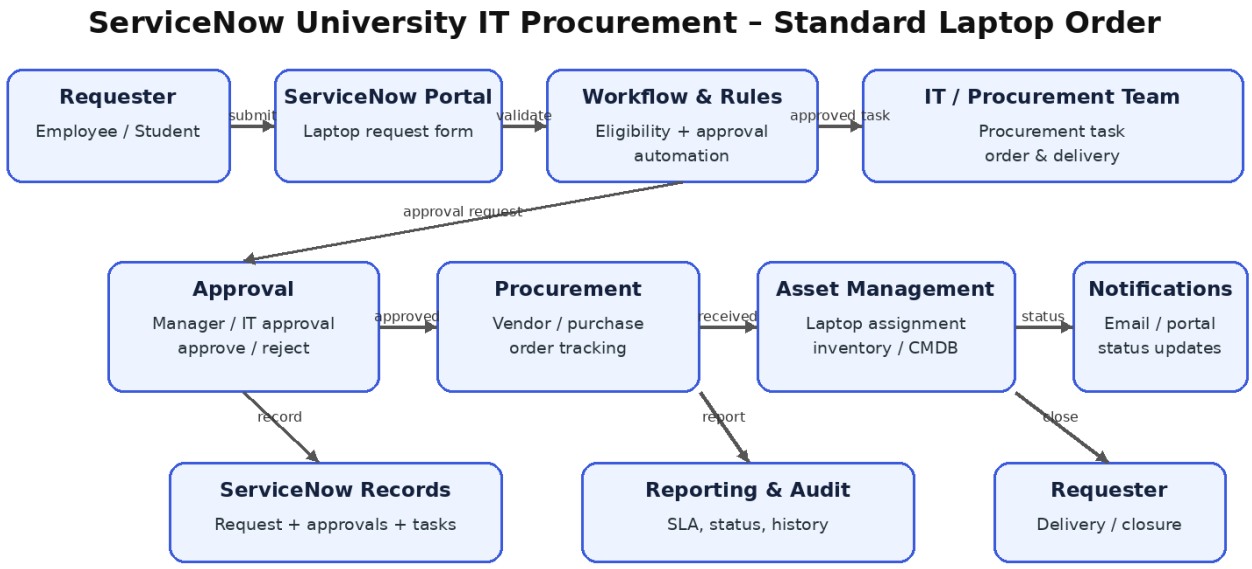


Figure 1: Solution architecture and data flow for the ServiceNow university IT procurement automation

Key Flow: Requester → ServiceNow request form → automated validation → approval → procurement task → laptop delivery/asset assignment → notification and closure. All major activities are recorded in ServiceNow for tracking, reporting, and audit purposes.

Technology / ServiceNow Components:

Component	Purpose
Service Catalog / Request Form	Collect standard laptop order details from the requester.
Flow Designer / Workflow	Automate validation, approvals, tasks, and status changes.
Business Rules / Data Policies	Enforce eligibility and required information.
CMDB / Asset Management	Track laptop inventory, assignment, and asset lifecycle.
Notifications	Send request, approval, procurement, delivery, and closure updates.
Reports / Dashboards	Monitor procurement volume, status, turnaround time, and audit history.

Reference: ServiceNow university IT procurement use case – solution architecture prepared for the project design phase.